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IoT Device Identification Base on Inter-flow Correlation Analysis Using Graph Neural Network

Chuang Peng[1], Xiaobin Tan[1,2], Peng Xie[1], Mengxiang Li[1], Hao Wang[1], Shuangwu Chen[1,2]
[1]University of Science and Technology of China, Hefei, China
[2]Institute of Artificial Intelligence, Hefei Comprehensive National Science Center, Hefei, China
Email: pengchuang@mail.ustc.edu.cn, xbtan@ustc.edu.cn, xp15002913199@gmail.com
limx2017@mail.ustc.edu.cn, wanghao1014@mail.ustc.edu.cn, chensw@ustc.edu.cn

***Abstract*—With the development of IoT, as one of the key technologies of IoT intelligence, IoT device identification is becoming increasingly important. However, the heterogeneity of devices and the sparsity of communication traffic data pose difficulties for IoT device identification. In this paper, we analyze the behavior and relationships between multiple flows of IoT traffic before concluding that there exist temporal correlations between multiple flows generated by one device. Based on this, an inter-flow temporal correlation graph (TCG) is constructed. A node in the TCG corresponds to a flow, and its features are the properties of the flow itself. An edge represents the temporal correlation between two flows. Then, taking into account the inter-flow temporal correlation, we propose a graph neural network-based IoT device identification method (TCG-DI). The effectiveness of TCG-DI is demonstrated through experiments based on open-source dataset and self-made dataset with high identification accuracy.**

***Index Terms*—IoT Device Identification, Encrypted Traffic Classification, Inter-flow Temporal Correlation, Graph Neural Network**

I. Introduction

Internet of Things (IoT) device identification is a technology that can identify and classify smart devices connected to the IoT. This technology can help organizations or enterprises identify and manage their IoT devices and ensure their security, which is one of the key technologies to achieve the intelligence of the IoT. In recent years, with the continuous development of IoT technology, its research and application prospects have become increasingly broad.

Different IoT devices will have different Quality of Service (QoS) requirements, such as network bandwidth or packet loss tolerance, due to differences in functionality. This heterogeneity makes the identification of IoT devices difficult. At present, IoT device identification mainly relies on network protocol analysis and machine learning technology [1], [2], through the analysis of the device's traffic protocol, traffic data, transmission behavior, and other information to extract the characteristics of which for device identification. However, some IoT devices generate very little intensive traffic, and the effectiveness of these methods is limited when applied to IoT devices because of the scarcity of communication.

Several research [3], [4] have shown that transforming network flows into graph-structured data is effective for network traffic identification. Deep learning on graphs is extended by Graph Neural Network (GNN) [5], which is well suited for handling graph-structured data. We find that for the traffic of IoT devices, there are correlations between multiple flows when viewed at the level of session flows. Therefore, we take a solution for IoT device identification: consider the correlations between flows and choose the GNN as the identification model.

In this paper, we construct an inter-flow temporal correlation graph (TCG) for network flows based on the observation that one IoT device usually generates multiple flows with inter-flow temporal correlation. Then, we propose an IoT device identification scheme based on the temporal correlation of flows using graph neural network (TCG-DI), in which the Graph Convolutional Network (GCN) [6] and Self-Attention Graph Pooling (SAGPool) [7], [8] are adopted to extract comprehensive representations of TCGs and thus enabling IoT device identification.

We summarize our main contributions as follows:

- We analyze the correlations between multiple flows generated by IoT devices, and then conclude that there exist exact temporal correlations between network flows generated by one device whether it is idle or active, which can be used for more accurate IoT device identification.
- We propose the TCG construction scheme based on inter-flow temporal correlation analysis, which can comprehensively characterize flows behavior pattern of a certain device by incorporating representative features of the flow itself and the temporal correlations between flows.
- We propose an IoT device identification method named TCG-DI based on the inter-flow temporal correlation of flows using graph neural network, in which the GCN and SAGPool are adopted to extract information to classify TCGs. We conduct a series of experiments based on open-source dataset and self-made dataset, and the results show that TCG-DI has excellent identification accuracy.

The rest of the paper is organized as follows. We first review the related work in Section II. Then, we make some preliminary analysis in Section III and present our method in Section IV. Next, we describe the datasets and evaluate the performance of TCG-DI through a series of experiments in Section V. We conclude this paper in Section VI.

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II. Related Work

QoS requirements for the same type of IoT devices are similar, e.g. smart cameras from the same manufacturer may have similar bandwidth requirements, meaning that a certain amount of labeled data can be used to train a classification model to achieve identification of IoT device brands and types. Through a series of works, Apthorpe et al. [9]–[12] demonstrated that an adversary can infer privacy activities from IoT device traffic, and proposed relevant defense strategies.

Meidan et al. [13] extracted about 300 packet-level and flow-level attributes from a single TCP flow, and used Random Forest to identify IoT devices. Sivanathan et al. [1] analyzed the traffic features of IoT devices and used 11 statistical features to achieve the identification of IoT devices based on Random Forest algorithm. In their subsequent work [2], server port number, DNS query, NTP query, and cipher suite are added to propose a two-stage IoT device identification method that sequentially goes through Bayesian classifier and Random Forest classifier. Marchal et al. [14] proposed AuDI for IoT device identification using unsupervised learning methods to model the periodic communication traffic of IoT devices. Pinheiro et al. [15] proposed a solution that uses packet length statistics such as statistical mean, the standard deviation, and the number of bytes transmitted over a one-second window from encrypted traffic to characterize the behavior of IoT devices, and used five machine learning algorithms to classify IoT devices.

With the successful application of deep learning in many fields, there have been many studies related to encrypted traffic identification using deep learning methods [16]–[18]. As for IoT device identification, Lopez-Martin et al. [19] combined CNN and RNN to achieve good results in IoT device identification. Ren et al. [20] proposed a deep learning-based model JCIDR that fuses numerical features and binary pictures of packets, which has 86%-95% accuracy in the joint classifications of devices and relation.

Additionally, there have been several attempts to identify IoT device events (such as bulbs turning on and off) from network traffic [21]–[23].

III. Analysis and Insight

In this section, we first introduce the threat model in this paper. Then, we investigate the traffic behavior of IoT devices. Finally, we interpret the insight of the proposed method.

A. Threat Model

We are concerned with leaking private information by analyzing network traffic from smart home devices, even though most of the traffic is encrypted. Currently, there are two types of adversaries: a WAN sniffer and a WiFi sniffer [23]. The WAN sniffer monitors network traffic communicated between the home router and the ISP network [9]–[12], which can examine the IP headers of all packets, but does not know the MAC address of the device. The WiFi sniffer monitors encrypted IEEE 802.11 communication, but there aren't many studies on it [12].

In this paper, we focus on the WAN sniffer. We assume that the adversary knows the type of smart home device they are monitoring. Therefore, they can train an IoT device identification model based on another device traffic of the same type and identify the traffic from the target smart home.

B. Traffic Behavior Analysis of IoT Devices

We analyze the behavior of IoT device traffic with data from CIC IoT Dataset 2022 [24]. In this paper, a flow refers to a session flow, also known as a bidirectional flow. We find that IoT devices generate periodic flows in their traffic whether they are in idle or active state because the devices send packets to servers to report their status at regular intervals. In addition, we find that many IoT devices generate more than one flows even in idle state.

Take the Amazon Alexa Echo Dot as an example, it generates about 15 TCP flows in half an hour. We observe the uplink flow of each TCP session flow, discrete each flow into a binary time series of one value sampled per second, which is used to indicate whether the flow sends one or more packets during the 1-second period, with a value of 1 at the corresponding location if yes and 0 otherwise. We find that the device generates two periodic TCP flows, and the active times of the two uplink flows are shown in Fig. 1a and Fig. 1b, where we can see that the flow with source port 56680 has a period of about 30 seconds, while the other has a period of about 90 seconds. Therefore, relying on a single flow to identify IoT devices is problematic.

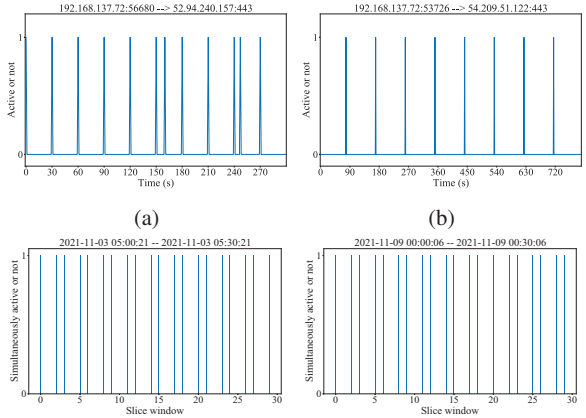


Fig. 1: Example of Amazon Alexa Echo Dot traffic flows

Existing research [3] has demonstrated that temporal correlation e.g., the active time correlation between flows can be used to describe the relationship between flows and thus help in the aggregation of information. The active time correlation between flows is to slice a period of time into several time windows and observe whether two flows are active simultaneously within the slice window, the more windows that are active simultaneously, the stronger the correlation between the two flows.

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We still take these two periodic flows as an example and take the flows from 05:00:21 to 05:30:21 on November 3, 2021, and 00:00:06 to 00:30:06 on November 9, 2021, each time period lasts half an hour, and the slice window length is set to 60 seconds, so each time period is divided into 30 slice windows, and then determine whether the two flows are active simultaneously in each slice window. As shown in Fig. 1c and Fig. 1d, the number of simultaneously active slice windows is the same, both are 20, even though the flows are extracted from different time periods of two days. This suggests that for IoT devices, the values obtained from the calculation of this temporal correlation will be more stable due to the presence of periodic flows, and thus can be used as features to construct behavior patterns of traffic.

C. Insight

Based on the previous analysis, we believe that an effective IoT device identification scheme should derive information not only from the features of individual flows, but also from the temporal correlations of multiple flows. Since IoT devices continuously generate TCP traffic, we implement IoT device identification primarily using TCP traffic in this paper.

Regarding the choice of data structure and classification model, due to the uncertain number of flows and the correlations between them, we came up with the idea of using graphs to represent multi-flow and their temporal correlations, using graph neural network for IoT device identification. With the ability to define relationships between multi-flows, graph can easily convert multiple flows into non-Euclidean space, allowing us to retain more original looks and characteristics without Euclidean distance constraints. Because graph neural network can accept data with an arbitrary number of flows as inputs, we can avoid data truncation and zero padding. In other words, all traffic flows generated by a single device within a period can be completely translated to a graph representation without losing important or adding redundant information, preserving data fidelity.

IV. The Proposed Method

In this section, we present the design details of the proposed GNN-based IoT device identification method named TCG-DI.

A. Overview

The overview of TCG-DI is presented in Fig. 2. In the graph generation process, we parse the TCP traffic packets sequence generated by one IoT device within 30 minutes into session flows according to the five-tuple (source IP address, source port, destination IP address, destination port, and transport layer protocol). For each flow, we obtain the periodic features and statistical features. Then, we construct the *Inter-flow Temporal Correlation Graph (TCG)* for the traffic. Therefore, the IoT device identification can be turned into a graph classification problem and TCGs are fed into GNNs to train a graph classifier. We employ GCN and a SAGPool layer to construct the main body of GNN model. It is worth noting that we do not extract information unique to TCP protocol traffic, so input traffic data can be augmented by integrating traffic of other protocols if necessary.

In the training stage, TCG-DI learns and trains IoT traffic end-to-end, and adjusts the parameters according to the training results. In the testing stage, the well-trained model is used to identify TCGs constructed by the traffic in the test dataset.

B. Inter-flow Temporal Correlation Graph

In this paper, we quantify the temporal correlation between flows and flows by calculating the active time cross-correlation [3], as shown in Eq. 1. For a flow and input time length d seconds, slicing it with a window of τ_{window} seconds obtains N slice windows. If a flow c_i has sent or received packets in the nth slice window, this flow is considered active in that slice window, i.e., $c_i[n] = 1$, otherwise $c_i[n] = 0$. Eq. 1 calculates the number of slice windows in which the flows c_i and c_j are simultaneously active. Then normalize the calculated cross-correlation, as shown in the Eq. 2.

$$c_i \otimes c_j = \sum_{n=1}^{N} c_i[n] \cdot c_j[n]. \tag{1}$$

$$(c_i \otimes c_j)_{norm} = \frac{\sum_{n=1}^{N} c_i[n] \cdot c_j[n]}{\sum_{n=1}^{N} max(c_i[n], c_j[n])}. \tag{2}$$

Based on the above cross-correlation, we construct the inter-flow temporal correlation graph, where each node represents a flow. Flows in TCG are connected by weighted edges, and the weight of each edge is determined by the value of the cross-correlation calculated by the two connected flows.

C. Flow Feature Selection

In the TCG, a node represents a certain session flow. Because node information will play a significant role in the GNN recognition result, it is critical to choose appropriate features to characterize the nodes. The node features we use consist of periodic features and statistical features.

1) Periodic Features: There have been some existing research works on period inference of traffic, however, they tend to divide traffic by protocols [14]. In IoT device traffic, flows based on a five-tuple cut are probably with exact tasks, so this rough division can cause problems for period inference work. We propose a period inference method based on flow motivated by existing study, which focuses on extracting periodic features by Discrete Fourier Transform (DFT) and autocorrelation. The specific steps to get periodic features from a session flow F are as follows:

- Take the uplink packet sequence of F, and convert it into a discrete binary sequence $y(t)$ of duration d seconds, which indicates whether the uplink flow contains one or more packets during 1 second, with a value of 1 at the corresponding position if yes, and 0 otherwise. Use DFT to convert $y(t)$ from the time domain to frequency domain and get $Y(f)$;
- Determine the maximum amplitude $Y(f_m) = max(Y(f))$ in the frequency domain, find $Y(f_c)$

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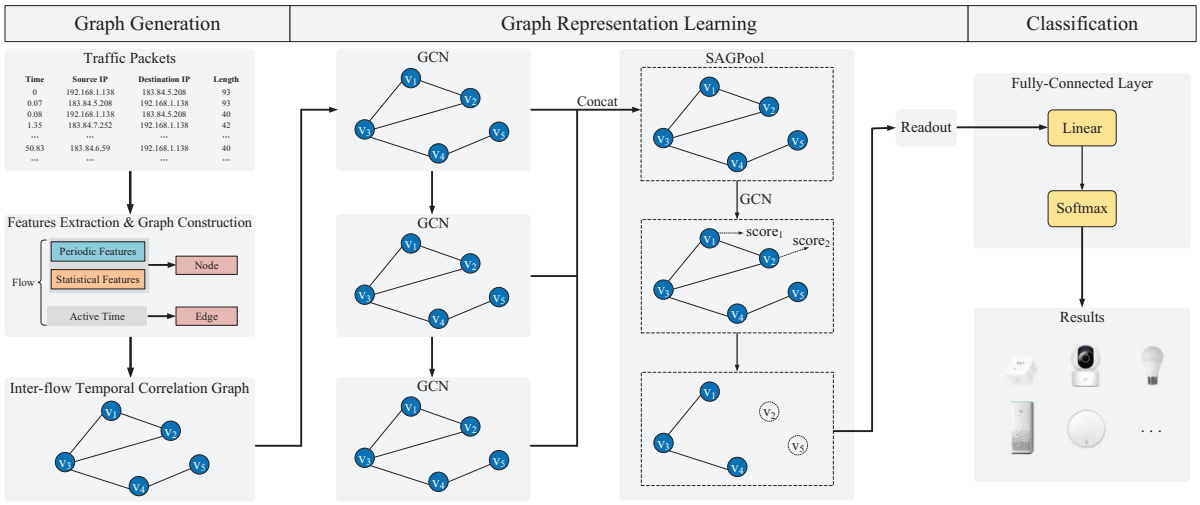


Fig. 2: The system overview of the TCG-DI

that satisfies the amplitude not less than $0.9Y(f_m)$ and $Y(f_c)$ is the local maximum of $Y(f)$, then f_c is the candidate frequency, i.e., the candidate period $T_c = \frac{d}{f_c}$ is obtained;

- To confirm and refine these candidate periods, the auto-correlation $R_{yy}(l)$ for $y(t)$ is calculated. If $R_{yy}(l)$ has a large value at l_c and is a local maximum, this means that $y(t)$ is likely to be periodic. For each candidate period T_c, if $R_{yy}(l)$ reaches a local maximum at l_i and l_i is within the range $[0.9T_c, 1.1T_c]$, then $T_i = l_i$ is considered a period of the flow;
- For each eligible period T_i, calculate r_i and rn_i using the Eq. 3 and Eq. 4, which are to measure the precision and stability of periodic T_i [14];
- For a flow, more than one eligible period may be computed, and for the sequence of eligible periods $Ts = [T_1, T_2, ..., T_m]$ and their corresponding $rs = [r_1, r_2, ..., r_m]$ and $rns = [rn_1, rn_2, ..., rn_m]$, the final periodic features are $[max(Ts), min(Ts), mean(rs), mean(rns)]$.

$$r_i = \frac{T_i \times R_{yy}(T_i)}{d}. \tag{3}$$

$$rn_i = \frac{T_i \times (R_{yy}(T_i - 1) + R_{yy}(T_i) + R_{yy}(T_i + 1))}{d}. \tag{4}$$

2) Statistical Features: According to [25], statistical features of flows can be used for encrypted traffic identification with good results. So we compute 18 statistical values for complete flow, uplink flow, and downlink flow respectively: minimum (min), maximum (max), mean ($mean$), median absolute deviation (mad), standard deviation (std), variance (var), skew ($skew$), kurtosis ($kurt$), percentile (from 10% to 90%) ($per1-per9$) and the number of packets (len). We can get 54 statistics for each flow and then use Random Forest method to rank the contribution of the statistical features. To reduce the parameters and model running time, we select M statistical features with the highest contribution as the statistical features of the nodes. The statistical features can be used as a complement to the periodic features to improve the classification accuracy of the model.

Therefore, a node in TCG can be represented by an F dimensional vector $\vec{h} = \{max(Ts), min(Ts), mean(rs), mean(rns), s_1, ..., s_M\}$, where the s_m represents a statistical feature, and $F = M + 4$.

D. GNN Architecture

Given a set of TCGs $\{G_1, ..., G_N\}$, and their labels $\{y_1, ..., y_N\}$, where the feature matrix of the graph G_i is X_i, GNNs aim to learn representation vector $\vec{H_G}$ that can predict the label of each TCG. We use GCN and SAGPool layer to construct the main body of GNNs in TCG-DI.

1) Graph Convolutional Layer: Graph convolutional network is a typical graph neural network that migrates convolution onto graph data and can aggregate node features.

The role of a graph convolutional neural network is essentially an update of graph node information. The graph convolutional layer is the only layer used in GCN. For a node on the graph, the neighbors of the node are first determined, and then the information of the neighboring nodes is aggregated to update its own information. Neighboring nodes can be determined by the adjacency matrix of the graph. According to matrix multiplication, the adjacency matrix of the graph is multiplied by the feature matrix of the graph, and the result obtained is that the new feature of each node is the sum of the features of its neighboring nodes, which gives the propagation rule of the graph convolution layer as shown in the Eq. 5, where $H^{(l)}$ is the node features matrix in the lth layer, $H^{(0)} = X$, $\tilde{A}$ is the adjacency matrix with added self-connections, $\tilde{A} = A + I$, I is the unit matrix, $\tilde{D}$ is the degree

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matrix of $\tilde{A}$, $W^{(l)}$ is the feature transformation matrix of the lth layer for the node features, and $\sigma(\cdot)$ denotes the activation function, such as the ReLU function.

$$H^{(l+1)} = \sigma(\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}H^{(l)}W^{(l)}). \tag{5}$$

GCN makes use of the structure and node information of graph data, and a single graph convolutional layer can aggregate information of neighboring nodes within one hop. Multiple graph convolutional layers can be used to extract high-level node representations. In TCG-DI, three tandem graph convolutional layers are used, and the output of each layer is spliced together.

2) Self-Attention Graph Pooling Layer: In order to focus more on important nodes, and weaken the influence of noisy nodes on recognition results, while reducing model parameters and improving training and testing efficiency, we use the self-attention graph pooling mechanism to pool the output of the GCN layers.

In TCG-DI, the SAGPool also uses the graph convolutional layer to learn a score for each node in the graph (i.e. the node feature dimension of the graph convolutional layer output is set to 1) and discards some nodes with low scores based on the ranking of all scores. This pooling approach borrows the idea of max pooling operation in CNN to filter out the more important information.

3) Readout Layer: Finally, the representation of the entire graph with K nodes is obtained through the readout layer:

$$\vec{H_G} = max_{i=1}^{K}\vec{h_i} \;||\; \frac{1}{K}\sum_{i=1}^{K}\vec{h_i}, \tag{6}$$

where $\vec{h_i}$ is the feature vector of node i. The readout layer operates the global pooling of graphs, which reduces the graphs with different numbers of nodes to the same dimension, and we implement it by splicing the global max pooling with the global average pooling.

4) Fully-Connected Layer: TCG-DI uses a linear function to perform linear transformation on the output data of the readout layer. Then, a dropout function is applied to avoid over-fitting. Finally, a softmax function is used to obtain the predicted probability vector indicating the likelihood that the input TCG belongs to each IoT device.

We leverage the cross entropy function as the loss function and adopt the Adam optimizer as optimizer in TCG-DI.

V. Evaluation

In this section, we are dedicated to evaluating the performance of TCG-DI. We first introduce our datasets, experiment setup, and metrics, then introduce the parameter setting of our model. Finally, we analyze the experiment results from different aspects.

A. Preliminary

1) Datasets: We first use the publicly available CIC IoT Dataset 2022 to optimize the parameters and validate the effectiveness of TCG-DI. Traffic in the CIC IoT Dataset 2022 is collected by wireshark and dumpcap, and the dataset contains IoT device traffic under two scenarios (Idle and Active). We use 7 days of Idle data and 10 days of Active data. In the Idle scenario, no one interacts with IoT devices throughout the lab, and in the Active scenario, all researchers are allowed to enter the lab at any time, and they may interact with devices, which passively or actively generate traffic for network communication. We select the traffic of 16 IoT devices in the dataset, covering common smart home devices such as smart speaker, smart plug, smart socket, smart camera, and floor sweeper.

We set up a testbed of IoT devices in our lab and collected traffic from 8 devices over two weeks, the dataset is named IoT8. Unlike CIC IoT Dataset 2022, IoT8 dataset does not intentionally differentiate between Idle and Active scenarios, users can interact with devices at any time, and automated settings are also made for some of these devices. IoT8 can further verify the robustness of TCG-DI.

2) Experiment Setup: There is no time overlap between each sample in the datasets, which are divided into a training set, a validation set, and a test set, with training set: validation set: test set = 3: 1:1. We use NetworkX to implement the graph construction and use PyTorch Geometric Library to implement graph neural network models.

3) Metrics: We use the four metrics $accuracy$, $precison$, $recall$, and $f1\text{-}score$ to measure the ability of the model to identify IoT devices.

B. Parameter Optimization

1) Selection of Statistical Features: More statistical features can provide more information, but features with greater randomness can bring interference to the deep learning model and lead to lower recognition accuracy, so the more useful features need to be selected. We used the Random Forest method to select the top 10 statistical features that contributed the most to the classification of the dataset, and the scores corresponding to each feature are shown in Fig. 3. In this figure, a-, u-, and d- denote the statistical features of the complete flow, the uplink flow, and the downlink flow, respectively. For example, $dlen$ means the number of packets of the downlink flow.

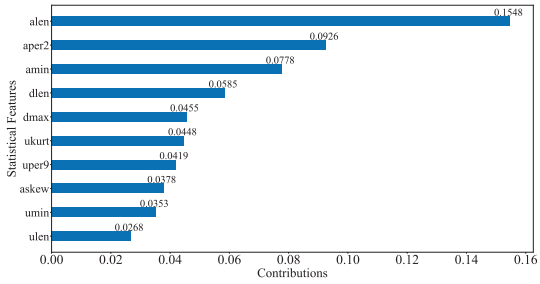


Fig. 3: Ranked contributions of individual statistical features

The length of periodic features is 4-dimensional, so the feature length of nodes in TCG is 14-dimensional.

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2) Hyperparameters of TCG-DI: We conduct an extensive search through hyperparameter space to make our model achieve its best performance. Table I shows the selected hyperparameters.

TABLE I: Hyperparameters selection for TCG-DI

Hyperparameters	Search Range	Values
Input Dimension	-	14
Hidden Dimension	[64...2048]	128
Learning Rate	[0.0005...0.01]	0.0005
Training Epochs	[10...100]	100
Batch Size	[16...256]	64
Pooling Ratio	[0.1...0.8]	0.8
Dropout Ratio	[0.1...0.8]	0.1

C. Experiment Results and Analysis of CIC IoT Dataset 2022

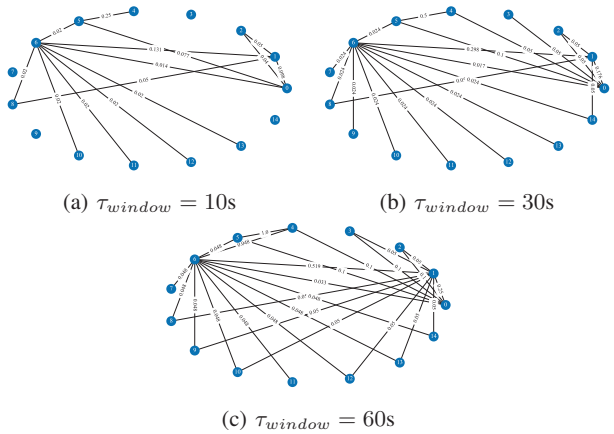
We used TCG-DI to compare with CCIT [1], a classic method, which uses sleep time, active volume, average packet size, mean rate, peak rate, active time, number of servers, number of protocols, unique DNS requests, DNS interval, and NTP interval as features to achieve the identification of IoT devices based on Random Forest algorithm.

For each IoT device, TCG is generated from 1800 seconds of TCP traffic. Experiments with TCG-DI are conducted using different parameters for Idle and Active data, containing different slice window lengths for computing correlation and different node features, and the results are shown in Table II.

1) Impact of Slice Window Length: We did experiments with slice window lengths of 60 seconds, 30 seconds, and 10 seconds, respectively, and the experiment results show that the window length of 10 seconds could achieve 99% accuracy with the best results for both scenarios of data. We consider that the reason for this result is that different window lengths correspond to different structures of temporal correlation graphs, which leads to the difference in identification results.

As shown in Fig. 4, we can see that both the number and the weight of the edges increase for Amazon Alexa Echo Dot TCGs with the increase of the slice window length. We believe that different slice window lengths bring different graph structures, and too long a window will lead to the appearance of edges with weights between less relevant flows, bringing redundant information to interfere with the identification.

2) Impact of Node Features: After determining the appropriate slice window time is 10 seconds, the node features are set to only periodic features (PF) and only statistical features (SF). As we can see from Table II, the accuracy of using only periodic features in the two scenarios is about 72% and 75%, respectively, which proves that the periodic features are useful, but they are not enough to be used as all features of the node. There are two reasons for this: first, the periodic features are calculated by DFT, and the error brought by the calculation is unavoidable, for example, the eligible period may not be obtained, which results in all nodes in a TCG having 0 vectors of features; second, as the number of IoT devices increases, there are bound to be IoT devices with



(a) $\tau_{window} = 10$s (b) $\tau_{window} = 30$s

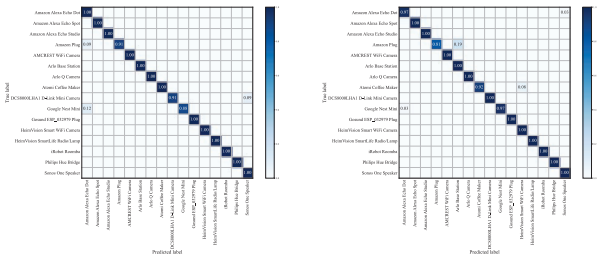
(c) $\tau_{window} = 60$s

Fig. 4: Amazon Alexa Echo Dot TCGs for different slice window lengths

similar periodic features. When the node features are only statistical features, an accuracy of about 96% can be achieved, which is relatively good, but still lower than the result of combining the two. The results show that the node features selected in this paper are more reasonable.

3) Comparison with CCIT: As shown in Table II, CCIT obtains fairly excellent performances with an accuracy of about 98%, but is still lower than TCG-DI with optimal parameters. We think that what limits the performance of CCIT is that it does not consider the correlations between flows and flows, even if the statistical features it uses are good enough. This result demonstrates the validity of the inter-flow correlation used by TCG-DI.

4) Identification for Individual IoT Devices: To further analyze the TCG-DI identification of each IoT device, the confusion matrix of identification results on both scenarios with slice window length of 10 seconds is plotted in this paper, as shown in Fig. 5.



(a) Idle scenario (b) Active scenario

Fig. 5: Confusion matrix for IoT device identification of CIC IoT Dataset 2022

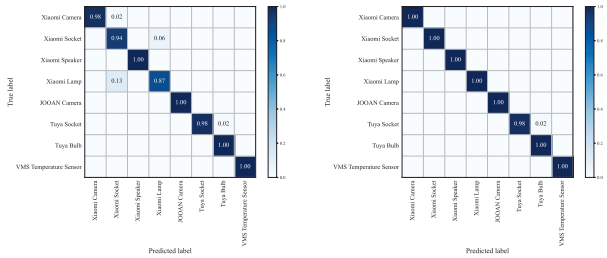
Fig. 5 shows that TCG-DI achieves 100% identification accuracy for most of the IoT devices in both the Idle and Active scenarios of the dataset. However, it can be seen that TCG-DI makes errors in the identification of Amazon Plug in both scenarios. By analyzing the raw traffic data, we find

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TABLE II: Experiment results of the CIC IoT Dataset 2022

Model	Idle				Active			
	Precision	**Recall**	**F1-score**	**Accuracy**	**Precision**	**Recall**	**F1-score**	**Accuracy**
CCIT	0.9821	0.9805	0.9810	0.9809	0.9865	0.9868	0.9864	0.9868
TCG-DI (PF)	0.7075	0.7201	0.6974	0.7201	0.7600	0.7566	0.7474	0.7566
TCG-DI (SF)	0.9737	0.9699	0.9704	0.9699	0.9701	0.9637	0.9647	0.9637
TCG-DI ($\tau_{window} = 60$s)	0.9743	0.9719	0.9715	0.9719	0.9850	0.9835	0.9837	0.9835
TCG-DI ($\tau_{window} = 30$s)	0.9870	0.9749	0.9861	0.9861	0.9898	0.9890	0.9892	0.9891
TCG-DI ($\tau_{window} = 10$s)	**0.9935**	**0.9931**	**0.9930**	**0.9931**	**0.9942**	**0.9939**	**0.9939**	**0.9939**

that the traffic generated by Amazon Plug within half an hour often contains only one TCP flow, which means that the TCG of Amazon Plug contains only one node, resulting in the inability to take advantage of the graph and graph neural network. Moreover, the period of Amazon Plug's flow is about 30 seconds, and Amazon Alexa Echo Dot also has flows with a period close to 30 seconds, which also means that some features of Amazon Plug are close to other devices, and we believe that the above reasons lead to the identification error. In addition, we also find that Google Nest Mini was incorrectly identified as Amazon Alexa Echo Dot in both scenarios, which we speculate is probably due to the large fluctuation of Google Nest Mini traffic. The traffic generated by Google Nest Mini varies greatly in different time periods, and it may generate 4 TCP flows or 32 TCP flows in 30 minutes.



(a) TCP flows only (b) TCP & UDP flows

Fig. 6: Confusion matrix for IoT device identification of IoT8

D. Experiment Results and Analysis of IoT8

In order to further test the identification capability and robustness of TCG-DI for IoT devices, experiments were conducted based on the self-made dataset IoT8, and the identification results are shown in Table III and Fig. 6.

We first did experiments on the IoT8 dataset using TCG-DI based on TCP flows only, the identification results are satisfactory for most of the devices, which also verifies that the TCG-DI has strong identification capability and robustness. However, it can be seen that Xiaomi Socket and Xiaomi Lamp are relatively easy to misidentify from each other. By viewing the raw traffic data of the two devices, we found that these two devices generate only one TCP flow most of the time, and the periodic features and statistical features of the flows are similar, which is a pain point of current IoT device identification, i.e., for different devices of the same brand, as they come from the same manufacturer, the traffic behaviors of the devices are likely to converge, which leads to the possibility of extremely similar flows and thus identification errors. For this type of problem, our approach is to add the information of UDP flows to the traffic identification model to further integrate the information of multiple flows and thus improve the ability of traffic identification. The experiment results in Table III and Fig. 6b prove that the identification accuracy for Xiaomi devices is greatly improved after adding the information of UDP traffic. As for the two Tuya devices, incorporating UDP traffic does not assist with identification because each device generates only one UDP flow and these two flows are almost identical.

Therefore, for TCG-DI, it is optional to include additional traffic information beyond TCP traffic to improve identification performance as needed, such as UDP traffic.

E. Summary

We conducted experiments to analyze the selection of node features and the selection of the slice window length in TCG-DI. We also analyzed the identification results for IoT devices of two datasets, and the results showed that the scheme proposed in this paper has good identification ability.

VI. Conclusion

In this paper, we first analyzed real-world IoT device network flow behavior and concluded that there exist temporal correlations between flows generated by an IoT device. Then, we constructed an inter-flow temporal correlation graph for IoT traffic flows. Finally, we proposed a novel IoT device identification method named TCG-DI based on inter-flow temporal correlation analysis using graph neural network. Experiment results showed that TCG-DI achieved excellent performance. In future work, we plan to expand the dataset, improve TCG-DI and evaluate its performance in large-scale scenarios.

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TABLE III: Experiment results of IoT8

Device	Manufacturer	TCP flows only			TCP & UDP flows		
		Precision	Recall	F1-score	Precision	Recall	F1-score
Smart Camera	Xiaomi	1.0000	0.9787	0.9892	1.0000	1.0000	1.0000
Smart Socket	Xiaomi	0.8627	0.9362	0.8980	1.0000	1.0000	1.0000
Smart Speaker	Xiaomi	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Smart Lamp	Xiaomi	0.9302	0.8696	0.8989	1.0000	1.0000	1.0000
Smart Camera	JOOAN	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Smart Socket	Tuya	1.0000	0.9787	0.9892	1.0000	0.9787	0.9892
Smart Bulb	Tuya	0.9792	1.0000	0.9895	0.9792	1.0000	0.9895
Smart Temperature and Humidity Sensor	VMS	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Accuracy			0.9707			0.9973	

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